

**Kindergarten
Ontario Curriculum
Foundations of Language and
Mathematics — Number Sense
and Operations**

The Counting Crew

Unit Overview: The Counting Crew

Grade: Kindergarten | **Strand:** Foundations of Language and Mathematics — Number Sense and Operations

Target Expectations: A6.1, A6.2, A6.3, A6.4, A6.5, A6.6

Learning Goal: Children will explore why numbers are used, count forward and backward, recognise and write numbers to 20, subitize quantities to 5, compose and decompose numbers to 10, and compare quantities as more, less, or equal.

Lesson	Lesson Title	Learning Goal
Day 1	What Are Numbers For?	I can find places where numbers are used in our classroom and tell why they matter.
Day 2	Forward and Backward	I can count forward and backward and tell what happens to the amount.
Day 3	Number Letters	I can read and write the numbers from 0 to 20.
Day 4	How Many Without Counting	I can look at a small group and say how many without counting.
Day 5	Building Numbers	I can build a number lots of different ways and compare two groups using more, less, or equal.

Lesson 1: What Are Numbers For?

Goal: Numbers do four jobs: count, label, measure, and tell time. They are everywhere in our classroom.

Expectation: A6.1

Learning Goal: "I can find places where numbers are used in our classroom and tell why they matter."

Materials: M1_counting_cubes, M5_counting_objects_set

1. Minds On (10 mins): Where Are the Numbers?

Today we welcome two new friends to our classroom: Mae and Theo, the Counting Crew chefs. They cook delicious snacks — but every recipe they make needs NUMBERS. They need YOUR help to find numbers all over the room.

Bring children to the carpet. Hold up the Mae illustration in one hand and the Theo illustration in the other. 'Friends, meet Mae and Theo. They are the Counting Crew chefs in our classroom kitchen. They are baking cookies today, and they need a lot of numbers.

2. Action (20 mins): The Number Hunt

- Whole group: introduce the Number Hunt clipboard.
- Send partner pairs out with one clipboard between them.
- Circulate and prompt with the diagnostic checklist.
- Pairs return to the carpet with their clipboards.

3. Consolidation (10 mins): Counting Crew Conversation

Discuss:

- Mae just asked us — could she bake cookies WITHOUT numbers?
- Theo wants to know — what was the most surprising number you found today?
- If our classroom did not have ANY numbers in it, what would be different?

Exit: Refrain chant 3x.

Lesson 2: Forward and Backward

Goal: Counting FORWARD makes amounts BIGGER; counting BACKWARD makes them SMALLER. Direction tells us.

Expectation: A6.2

Learning Goal: "I can count forward and backward and tell what happens to the amount."

Materials: M7_counting_path, M1_counting_cubes, M5_counting_objects_set

1. Minds On (10 mins): Mae's Recess Countdown

Mae and Theo are baking apple muffins, but they keep peeking at the timer because RECESS is coming. Today the Counting Crew will teach us a special chef trick: counting BACKWARD when we are waiting for something exciting.

Bring children to the carpet. Hold up Mae and Theo. 'Friends, Mae and Theo started baking muffins twelve minutes ago. They keep asking — how long until recess?' Hold up five paper apples in a basket. 'Yesterday we counted UP — one apple, two apples, three apples — to find HOW MANY. The amount got BIGGER each time. Watch.'

2. Action (20 mins): The Counting Path — Forward and Back

- Whole group at the Counting Path.
- Partner pairs take the Counting Path two at a time (others watch and chant).
- Mid-Action twist: 'mystery start'.

3. Consolidation (10 mins): Bigger or Smaller? Counting Crew Conversation

Discuss:

- Mae just asked us — when she counts the muffins coming OUT of the oven, is that forward or backward?
- Theo wants to know — if our timer says 5, 4, 3, 2, 1 ... what is happening to the time we have left?
- Show me with your hands: was it BIGGER or SMALLER when you walked toward the red arrow on the Counting Path?

Exit: Refrain chant 3x.

Lesson 3: Number Letters

Goal: Each number has its own written form. We can read and write 0-20. Teens (11-19) need TWO digits.

Expectation: A6.3

Learning Goal: "I can read and write the numbers from 0 to 20."

Materials: M4_number_cards, M1_counting_cubes, M5_counting_objects_set

1. Minds On (10 mins): The Mixed-Up Recipe Cards

Mae and Theo were writing down a new cookie recipe — but a gust of wind blew their number cards everywhere! They need OUR help to put the numbers back in order from 0 to 20 so they don't accidentally bake 30 cookies instead of 13.

Bring children to the carpet near the yarn 'clothesline'. Hold up a basket of number cards 0-20, all jumbled. 'Look, friends — Mae and Theo's recipe cards got blown out of order. The clothesline is empty. We need to peg every number from zero up to twenty, in the right order.

2. Action (20 mins): Number Stations — Match, Order, Trace

- Set up three rotating stations.
- Rotate every 6 minutes.
- Final 4 minutes: pull the whole class back to the long carpet clothesline.

3. Consolidation (10 mins): Counting Crew Conversation — Tricky Teens

Discuss:

- Mae just asked us — when she writes 13, what TWO digits does she use?
- Theo is mixing up 14 and 40.
- If I say a number out loud and you have to find it on the clothesline, what is the FIRST clue you look for?

Exit: Refrain chant 3x.

Lesson 4: How Many Without Counting

Goal: Sometimes we know HOW MANY just by LOOKING — that's subitizing. Works for 1-5. Past five we count.

Expectation: A6.4

Learning Goal: "I can look at a small group and say how many without counting."

Materials: M3_dot_cards, M2_ten_frames, M1_counting_cubes

1. Minds On (10 mins): Buddy's Paw-Print Flash

Buddy the puppy is back, and he has a NEW trick. Today he is going to show us how to spot HOW MANY paw prints he leaves WITHOUT counting them. This trick is called subitizing, and only super-fast eyes can do it.

Bring children to the carpet. Hold up Buddy and a small stack of paw-print cards. 'Friends — Buddy walked through a puddle and left paw prints. I am going to FLASH a card very fast, and your job is to say how many paw prints you saw without counting one by one.'

2. Action (20 mins): Subitize, Build, Sort — Partner Centres

- Whole-group warm-up flash with the dot cards in pocket-chart.
- Move to partner centres in two rotations of 7 minutes each.
- Whole class returns to the carpet for the final 4-minute 'subitize-and-tell' game.

3. Consolidation (10 mins): Counting Crew Conversation — Eyes That Know

Discuss:

- Buddy wants to know — what is the BIGGEST number our eyes can just KNOW without counting?
- Mae just asked us — when she sees a dice and rolls a 4, does she count the dots or just KNOW?
- Show me with your hands: ONE big group of 6 versus TWO little groups of 3 — which is easier to subitize?

Exit: Refrain chant 3x.

Lesson 5: Building Numbers

Goal: Two small groups can compose into one bigger, or decompose. We compare with MORE, LESS, or EQUAL.

Expectation: A6.5, A6.6

Learning Goal: "I can build a number lots of different ways and compare two groups using more, less, or equal."

Materials: M1_counting_cubes, M2_ten_frames, M6_compare_strip, M5_counting_objects_set

1. Minds On (10 mins): Buddy's Bone Plates

Buddy has TWO plates of bones — he wants the plate with MORE. Which one is it?

Bring children to the carpet. Hold up Buddy and two plates: 'Plate A has 6 bones. Plate B has 4 bones. Buddy wants MORE — which plate?' Affirm: 'Six is MORE than four. Four is LESS than six.' Lay out MORE / LESS / EQUAL cards.

2. Action (20 mins): Building 7 Lots of Ways, Then Comparing Plates

- Whole-group compose demo with target number 7.
- Partner compose station.
- Switch to compare station with cookie plates and the comparison strip.
- Whole-class debrief.

3. Consolidation (10 mins): Counting Crew Conversation — Same Number, Many Ways

Discuss:

- Mae just asked us — when she has 4 cookies and Theo has 4 cookies, what word do we use?
- Theo wants to know — if you build 6 with three reds and three yellows, can your partner build 6 a different way?
- Buddy got two plates: one with 5 bones, one with 3.
- When is it useful in real life to know if you have MORE or LESS? Think about lunch, recess, sharing.

Exit: Each child takes one number card 1-10 from the basket as they walk out and shows the teacher TWO ways to build that number with their fingers (e.g., for 5: shows 2+3, then 4+1).

Name: _____

Date: _____

Where Did You See a Number?

Learning Goal: I can find places where numbers are used in our classroom and tell why they matter.

Instructions for the Student:

1. Draw or write where you saw a COUNTING number today.
2. Cut out the picture cards at the bottom.
3. Fill in the blanks.

Sort the cards

COUNTING	LABELS	MEASURE	TIME

Name: _____

Date: _____

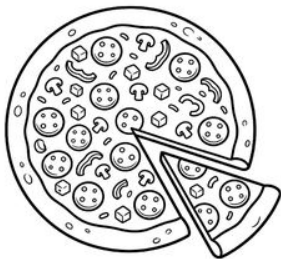
My Counting Path: Big or Small

Learning Goal: I can count forward and backward and tell what happens to the amount.

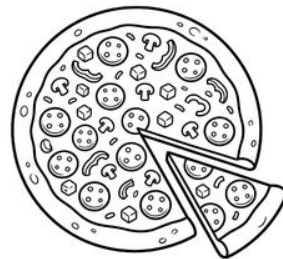
Instructions for the Student:

1. Trace the green arrow.
2. Trace the red arrow.
3. Draw three muffins on the tray.

Mae's Muffin Tray



BEFORE: 5 muffins



AFTER: 2 muffins

Name: _____

Date: _____

Match, Trace, Write Numbers

Learning Goal: I can read and write the numbers from 0 to 20.

Instructions for the Student:

1. Match each number card on the left to the matching group of objects on the right.
2. Trace the dotted numerals 0 to 10 on the top row.
3. Trace each teen numeral 11, 12, 13, 14, 15.

Teen visual: 15



+ 5

= 15

Name: _____

Date: _____

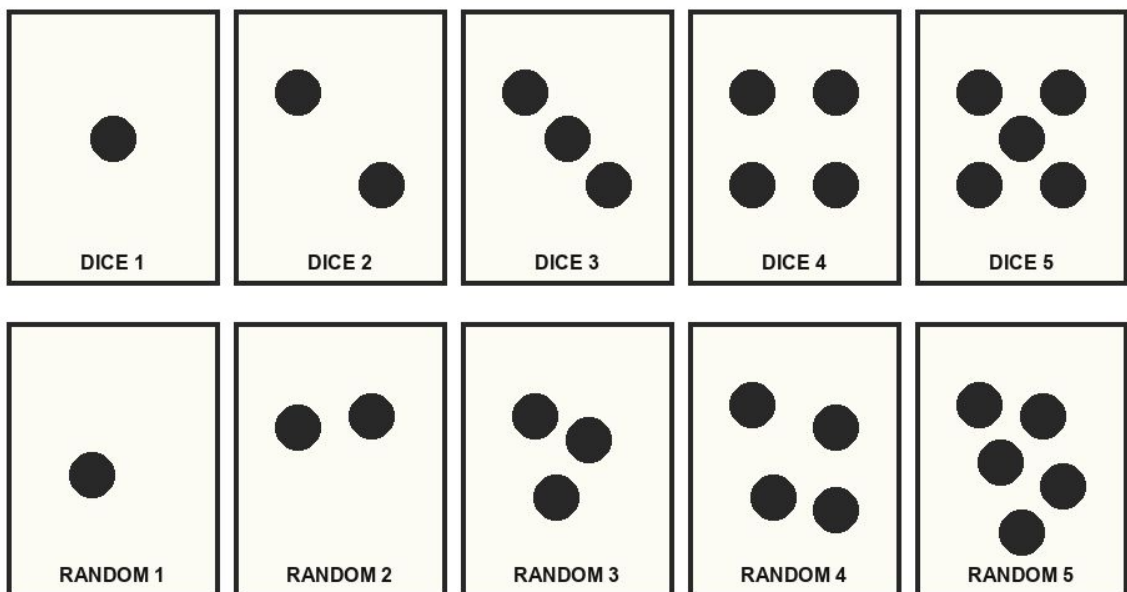
Subitize: How Many at a Glance

Learning Goal: I can look at a small group and say how many without counting.

Instructions for the Student:

1. Look at each dot card for a moment.
2. Cut out the cards at the bottom.
3. Buddy left some paw prints in different places.

Subitizing dot cards



Name: _____

Date: _____

Build 7 and Compare Plates

Learning Goal: I can build a number lots of different ways and compare two groups using more, less, or equal.

Instructions for the Student:

1. Build the number 7 in TWO different ways using red and yellow cubes.
2. Look at Plate A and Plate B.
3. Draw your own two plates of timbits.

Draw your own plates

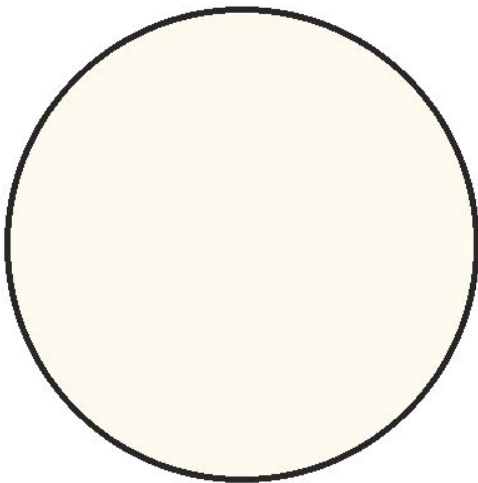


Plate A

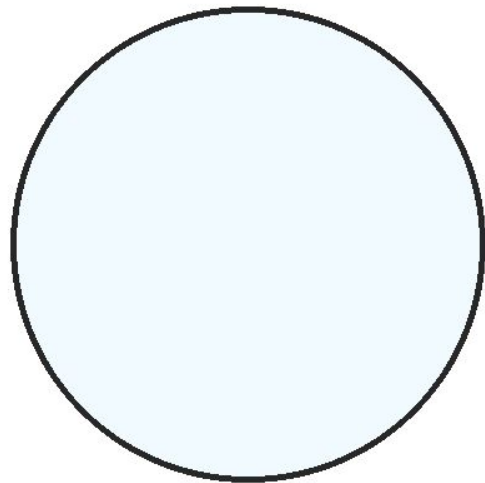


Plate B

Cube Sorting Mat

(M1_counting_cubes)

Linking snap-cubes are the core hands-on manipulative for this unit, used in every lesson by every child to count, compose, decompose, and compare quantities up to 20.

Cube Sorting Mat

COUNT HERE	GROUP A	GROUP B

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

USE IN CLASS

Quantity: 1 mat per child

Lessons: 1, 2, 3, 4, 5

Prep: ~25 min

TEACHER PREP

1. Print 1 Cube Sorting Mat per child (20 pages for a class of 20).

2. Laminate every sheet.

3. Confirm you have ~30 linking snap-cubes per child (not printed; classroom purchase).

4. Stack the laminated mats in the math centre.

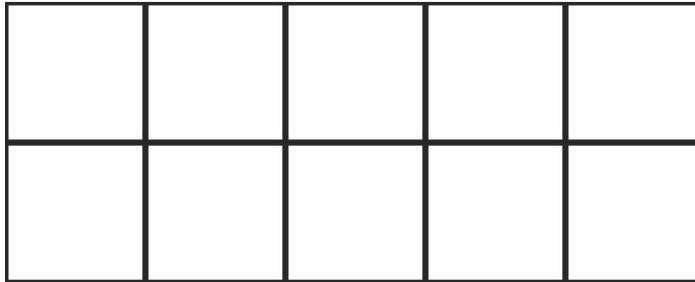
5. Optional colour: print COUNT HERE in blue and GROUP squares in yellow.

Ten-Frame Mats

(M2_ten_frames)

Laminated 2×5 grids that children fill with cubes, counters, or sticker dots to model quantities up to 10.

Ten-Frame Mat



Ten-Frame: 2 × 5 cells

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

USE IN CLASS

Quantity: 2 mats per child

Lessons: 4, 5

Prep: ~20 min

TEACHER PREP

1. Print 1 page per child.

2. Laminate before cutting so children can use dry-erase markers on the cells.

3. Cut each page along the solid horizontal line into 2 single ten-frame strips.

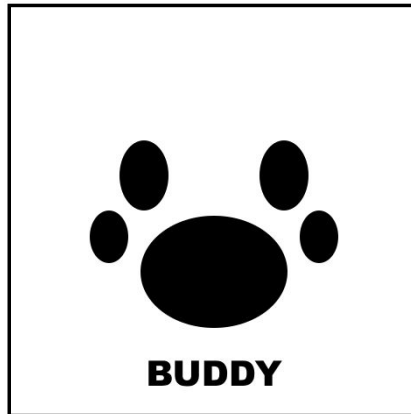
4. Distribute 2 mats per child before Lesson 4.

5. Pair each child's mats with a small bag of 10 round counters (or M1 cubes).

Subitizing Dot Cards

(M3_dot_cards)

A 10-card flash set the teacher uses for instant-recognition subitizing in Lesson 4.



PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 2

Laminate: yes

USE IN CLASS

Quantity: 1 deck

Lessons: 4

Prep: ~15 min

TEACHER PREP

1. Print 1 set (2 pages) for the teacher.

2. Laminate both pages before cutting so cards survive daily flashing.

3. Cut along the cut-lines into 10 individual cards (5 standard + 5 random).

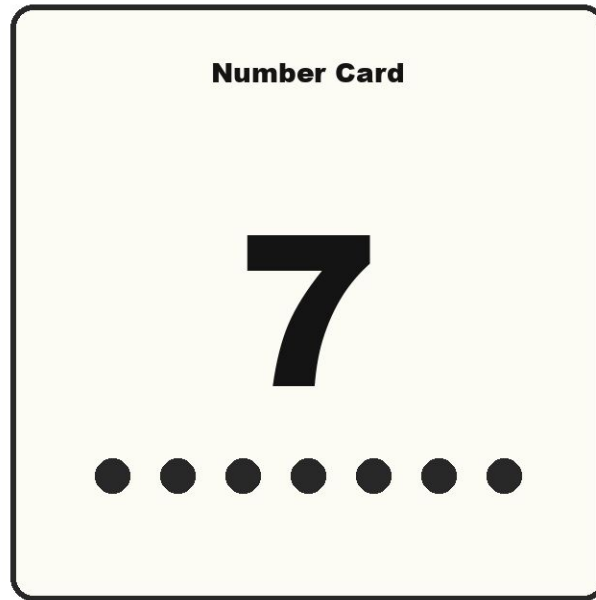
4. Store the deck in a labelled envelope.

5. Optional: punch a hole in each card and bind on a ring so cards stay together.

Number Cards 0-20

(M4_number_cards)

A 21-card numeral set printed letter-sized, used by children to sequence numbers, match a numeral to a quantity, write the missing number in a row, and identify the tricky teen numbers.



PRINT SPECS

Size: letter portrait
Colour: bw
Pages per set: 6
Laminate: yes

USE IN CLASS

Quantity: 1 class set
Lessons: 3, 5
Prep: ~30 min

TEACHER PREP

1. Print all 6 pages (1 class set).
2. Laminate every page before cutting so they survive a year of K hands.
3. Cut into 24 individual cards (numerals 0-20 + 3 Mystery cards).
4. Sort the 0-20 cards in order and bind on a ring.
5. Place at the Number-Sort station for Lesson 3 and the Compare station for Lesson 5.

...

Object Sorting Tray

(M5_counting_objects_set)

Real classroom objects (apples, water bottles, books, blocks, paper bears) are the diagnostic backbone of Lesson 1 — children count them around the room and sort them by purpose.

Object Sorting Tray

APPLES	BOTTLES	BOOKS	BLOCKS	BEARS

PRINT SPECS

Size: letter landscape

Colour: bw

Pages per set: 1

Laminate: yes

USE IN CLASS

Quantity: 1 tray per child

Lessons: 1, 2, 3, 5

Prep: ~25 min

TEACHER PREP

1. Print 1 sorting tray per child (20 pages for a class of 20).

2. Laminate every sheet.

3. Confirm you have ~10 of each real object: apples, water bottles, books, blocks, bears.

4. Stack laminated trays at the Counting Centre.

5. Optional colour: per-zone icons can be printed in colour.

More / Less / Equal Strip

(M6_compare_strip)

The summative manipulative for Lesson 5: a printable strip with two parallel rows where children align two groups of cubes (or counters) one-to-one and visually decide MORE, LESS, or EQUAL.

GROUP A	GROUP B

MORE

LESS

EQUAL

PRINT SPECS

Size: letter landscape
Colour: bw
Pages per set: 1
Laminate: yes

USE IN CLASS

Quantity: 1 strip per child
Lessons: 5
Prep: ~15 min

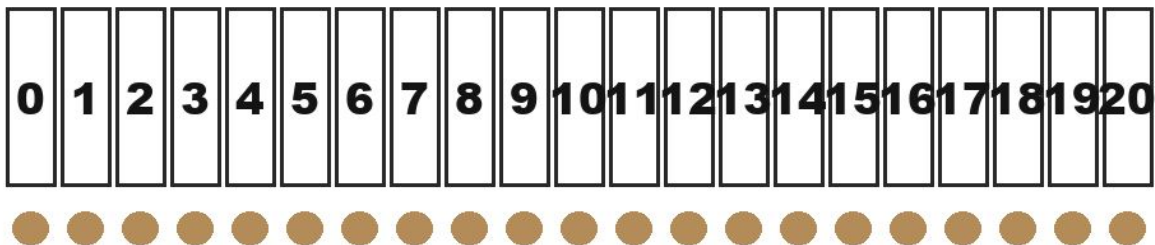
TEACHER PREP

1. Print 1 page per pair.
2. Laminate before cutting so children can use a dry-erase marker on MORE/LESS/EQUAL.
3. Cut along the horizontal cut-line into 2 single comparison strips per page.
4. Distribute 1 strip per child before Lesson 5 with ~10 cubes for each group.
5. Optional: provide dry-erase markers so children can circle their answer.

Counting Path Floor Mat

(M7_counting_path)

A long numbered floor mat that children physically walk forward (count up) or backward (count down) one number at a time.



Counting forward from 0 to 20

PRINT SPECS

Size: letter portrait
Colour: bw
Pages per set: 11
Laminate: yes

USE IN CLASS

Quantity: 1 class mat
Lessons: 2
Prep: ~45 min

TEACHER PREP

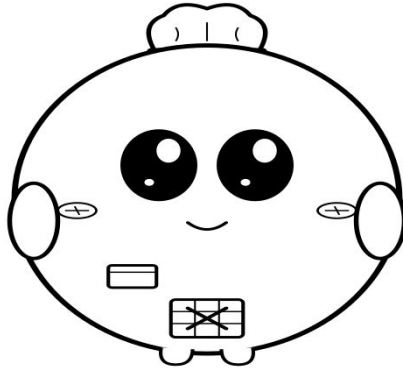
1. Print all 11 pages on cardstock if available (otherwise paper + heavy lamination).
2. Laminate every page front and back.
3. Cut every page along the horizontal cut-line into 22 single-tile sheets.
4. Lay tiles in a long line on the carpet from 0 to 20.
5. Roll up tiles after each lesson and store in a labelled tube.

...

Mae the Lead Counter Puppet

(char_mae_puppet)

A printable Mae illustration that the teacher mounts on a popsicle stick and uses as a recurring class character throughout all 5 lessons.



Mae the Counting Crew Chef

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

USE IN CLASS

Quantity: 1 puppet for the teacher

Lessons: 1, 2, 3, 4, 5

Prep: ~10 min

TEACHER PREP

1. Print 1 page on cardstock if possible (or paper + laminate for durability).

2. Cut around the outside of Mae's silhouette including the bottom tab.

3. Glue or tape a popsicle stick behind the bottom tab so Mae can be held up.

4. Optional: laminate before cutting for years of reuse.

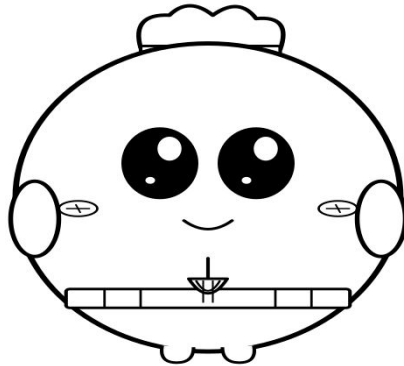
5. Store flat in a folder so the puppet does not crease.

...

Theo Partner Puppet

(char_theo_puppet)

A printable Theo illustration mounted on a popsicle stick, used as Mae's pair partner across all 5 lessons.



Theo the Counting Crew Chef

PRINT SPECS

Size: letter portrait
Colour: bw
Pages per set: 1
Laminate: yes

USE IN CLASS

Quantity: 1 puppet for the teacher
Lessons: 1, 2, 3, 4, 5
Prep: ~10 min

TEACHER PREP

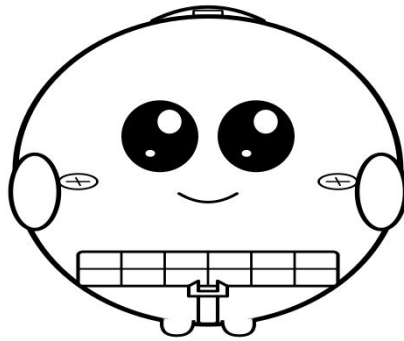
1. Print 1 page on cardstock if possible (or paper + laminate for durability).
2. Cut around the outside of Theo's silhouette including the bottom tab.
3. Glue or tape a popsicle stick behind the bottom tab so Theo can be held up.
4. Optional: laminate before cutting for years of reuse.
5. Store flat in a folder next to the Mae puppet so they always travel together.

...

Buddy Companion Puppet

(char_buddy_puppet)

A printable Buddy-the-puppy illustration mounted on a popsicle stick.



Buddy the Puppy

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

USE IN CLASS

Quantity: 1 puppet for the teacher

Lessons: 1, 4, 5

Prep: ~10 min

TEACHER PREP

1. Print 1 page on cardstock if possible (or paper + laminate for durability).

2. Cut around the outside of Buddy's silhouette including the bottom tab.

3. Glue or tape a popsicle stick behind the bottom tab so Buddy can be held up.

4. Optional: laminate before cutting for years of reuse.

5. Store flat in a folder; bring out for every lesson alongside Mae and Theo.

...

Name: _____

Date: _____

Summative Assessment Rubric

Expectation	Level 1 (Beginning)	Level 2 (Developing)	Level 3 (Achieving)	Level 4 (Exceeding)
A6.1 explore what numbers are used for	With significant adult prompting, names one place where a number appears in the room. Cannot yet describe what the number is doing.	Names two or three numbers in the classroom and pairs each with the right job (count, label, measure, time) when prompted by the anchor chart.	Independently finds at least three numbers around the room and explains the job of each in child language (e.g., 'this one labels the room').	Notices numbers spontaneously beyond the lesson (on lunch boxes, at home, on the way to recess) and articulates that numbers are used for many
A6.2 understand when counting forward, the quantity gets bigger and when counting backward, the quantity gets	Counts forward to 5 or 10 by rote but cannot count backward. Treats 'counting' as one direction only and may invert the bigger/smaller language.	Counts forward and backward with adult support. Articulates 'bigger' or 'smaller' when prompted but mixes them up some of the time.	Counts forward and backward independently across the 0–20 range and correctly says 'the number got bigger' or 'smaller' as the direction changes.	Counts backward from any starting point inside 0–20, applies bigger/smaller language unprompted, and notices the same pattern in real-life moments
A6.3 identify and write numbers to 20	Recognises a few digits 0–10 and may write them with reversals. Confuses teens with decades (e.g., reads 13 as 30) consistently.	Reads and writes 0–10 with occasional reversals. Reads teens correctly with anchor-chart support but may still flip 13/14/15 in writing.	Reads and writes 0–20 independently. Uses the tens-stick model to disambiguate teens from decades when needed (e.g., points to the 1 and says 'one	Reads, writes, and orders 0–20 fluently. Begins to write 21–30 unsupported and recognises the place-value pattern that each decade is a count of
A6.4 subitize quantities to 5	Recounts every dot-card from one each time. Cannot yet recognise small quantities (1–3) without counting.	Subitizes 1, 2, and 3 in standard arrangements but resorts to counting for 4 and 5 or for non-standard arrangements.	Subitizes 1–5 in standard (dice) arrangements without counting. Begins to articulate how they saw it (e.g., 'I saw two and two more').	Subitizes 1–5 in random arrangements and starts to subitize small groups beyond 5 by part-part-whole reasoning ('I saw a 3 and a 3 so it must be 6').
A6.5 compose and decompose quantities to 10	Builds a target quantity in only one way and cannot yet break it into smaller parts. May rebuild the target from scratch each time the teacher asks	Composes a target quantity in two different ways (e.g., $7 = 5+2$ and $7 = 4+3$) with adult prompting and the help of the two-colour cube split.	Independently composes a target quantity to 10 in at least two different ways and decomposes a given group into two smaller groups using the	Composes a quantity to 10 in three or more different ways and articulates a part-part-whole rule ('seven is six and one, OR five and two, OR four and
A6.6 understand the concept of more than, the same (equal), or less than	Compares two groups by appearance only ('the long one has more'). Does not yet line objects up to	Compares two groups by lining them up on the comparison strip with adult prompting. Uses 'more' and 'less' but	Lines up two groups on the comparison strip independently and correctly says which group has MORE	Compares two groups without needing to line them up — uses subitizing or counting to determine

Junior Counting Crew Certificate

This certificate is proudly presented to:

For successfully completing the 5-Day Counting Crew unit and demonstrating the skills of a Counting Crew chef. Mae and Theo, and Buddy the puppy, congratulate the student and welcome them to the next adventure in numbers.

By completing this unit, this student has demonstrated:

- Spotting numbers around the classroom and naming what each number is for
- Counting forward to 20 and backward from 20
- Reading and writing the numbers 0 to 20 — including the tricky teens
- Subitizing groups of 1 to 5 without counting

Marketplace Listing

A 5-day Ontario Kindergarten unit on Number Sense and Operations (A6.1–A6.6), framed around Mae and Theo, two child chefs, and Buddy the puppy.

Price: \$9.99 CAD

Pages: ~38

Classroom time: 200 min

Teacher prep: ~95 min

Top tags: number sense, counting, subitizing, kindergarten, Ontario, Ontario curriculum, math, mathematics

What's included:

- 5 fully-scripted lesson plans (40–45 minutes each, ~205 minutes total of teaching time)
- 5 student worksheets aligned to each lesson, with name/date fields and child-voice learning goals
- 1 mid-unit formative assessment ('Numerals 0–20 Check-In') administered between Lesson 3 and Lesson 4
- 1 end-of-unit reflection sheet with 5 prompts for student self-assessment
- 10 printable teacher manipulatives: counting cube storage mat, ten-frame mats, Buddy's paw-print subitizing dot cards, number cards 0–20, real-world counting...
- 4 class observation trackers: 1 diagnostic (Lesson 1) plus 3 formative (Lessons 2, 3, 4)
- 1 six-level summative rubric covering all six expectations (A6.1–A6.6)
- 1 summative administration guide tying the Lesson 5 worksheet, the lesson, and the rubric together for the teacher
- Junior Counting Crew certificate (one per child) for the closing ceremony, featuring Mae, Theo, and Buddy
- Detailed expanded walkthrough for the keystone lesson (Lesson 3 — Number Letters), including the tens-stick recovery for the 'thirteen vs thirty' confusion,...
- Differentiation tracks built into every lesson: emerging learners, extending learners, ELL/IEP, and movement-first learners